

Extraction of genomic DNA from tissue culture cells

Risk assessment

- Work with human-derived material or transgenic cell lines (BSL-2)
- ▷ Wear gloves, safety glasses, lab coat



Reviewed: Nov 11, 2023

This is a bench card. Full protocol available online.

>> Preparation of cell lysates

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|---|--|
| ☐  R0040 7 U/μL RNase A | ☐  R0142 Cell lysis buffer, 1 mL |
| ☐ 0.6–0.8 U/μL Proteinase K | ☐ Buffer PB (Binding buffer), pH 4.8 |

- (1.) Prepare 10 μL proteinase K in a sterile microcentrifuge tube.
- (2.) Harvest 1×10^6 – 5×10^6 cultured cells by trypsinization or equivalent method. Wash pellet in PBS. 
- (3.) Resuspend pellet in 200 μL PBS and transfer to tube containing proteinase K.
- (4.) *Optional:* Add 4 μL RNase A. Mix briefly. Incubate 2 min at room temperature.
- (5.) Add 200 μL cell lysis buffer. Vortex to obtain a homogenous solution.
- (6.) Incubate at 55 °C for 10 min to complete protein digestion.
- (7.) Add 200 μL Buffer PB. Mix thoroughly.

>> DNA purification

- (1.) Apply the lysate on a silica membrane spin column. Centrifuge at $11\,000 \times g$ for 1 min or use a vacuum manifold.
- (2.) Wash the silica membrane with 650 μL Buffer PE.
- (3.) Repeat the wash step once more.
- (4.) Dry membrane by centrifugation at $11\,000 \times g$ for 2 min.
- (5.) Place the spin column into a clean microcentrifuge tube.
- (6.) Apply 25–200 μL 5 mM Tris pH 8.0. Incubate 1 min. Elute by centrifugation at $11\,000 \times g$ for 1 min. 

Storage of genomic DNA

- (1.) Store at 4 °C or at room temperature for up to six months under slightly basic conditions (pH 8.0) or at –20 °C for long-term storage.

Quality assurance: Check DNA concentration after storage at 4 °C or room temperature to compensate for evaporation.



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